ModelIt!

Student Activity Pack

Can the Ocean Power Entire Cities?

Modeling Ocean Thermal Energy Conversion and Marine Renewable Power

Grade 10L3 | NGSS HS-PS3-2, HS-ESS3-1 | Energy and Matter

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 How do you think the temperature of the ocean changes as you go deeper?

2 Could you use a temperature difference to generate electricity? How might that work?

3 Draw what you think a power plant that runs on warm and cold water would look like.

4 Why do you think ocean energy is not widely used if the ocean covers 70% of the Earth?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Surface Water Temperature | Deep Water Temperature | Temperature Differential | Fluid Flow Rate | Turbine Efficiency | Energy Conversion | Environmental Impact | Infrastructure Cost | Power Output

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Tropical Optimal

28 degrees Celsius surface, 4 degrees Celsius deep, 24 degree differential

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Reduced Differential

Temperature differential drops to 15 degrees Celsius

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Maximum Scale

Fluid flow rate increased to maximum platform capacity

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Desalinated Water Output — OTEC systems can produce fresh water as a byproduct by condensing evaporated seawater — a valuable co-product for island communities with limited freshwater resources

Cold Water Agriculture — The cold deep water discharge can be used for air conditioning and cold-water agriculture (growing temperate crops in tropical climates), improving the economic viability of the OTEC system

Biofouling Rate — The rate at which marine organisms (barnacles, algae, biofilm) colonize heat exchanger surfaces, reducing thermal transfer efficiency and requiring periodic cleaning or anti-fouling treatment

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

The ocean absorbs more solar energy in one day than humanity uses in a year — can we tap this enormous reservoir to power civilization? Use evidence from your simulation to support your answers.

1. **How does the Carnot efficiency limit fundamentally constrain OTEC performance, and why is this different from solar or wind energy constraints?**

2. **What does your model reveal about the relationship between temperature differential and economic viability?**

3. **Why is the 24/7 baseload nature of OTEC valuable even though its per-kWh cost is higher than solar or wind?**

4. **How would climate change (warming surface waters, potentially changing deep water temperatures) affect OTEC performance over the next century?**

5. **What are the ethical considerations of potentially disrupting marine ecosystems to generate clean energy?**

8 STEM Challenge: Design an OTEC-Powered Island Energy System

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YOUR ENGINEERING MISSION

Design a comprehensive OTEC-based energy system for a tropical island nation that currently depends entirely on imported diesel fuel for electricity.

THE SCENARIO

A small Pacific island nation of 50,000 people currently generates all electricity from imported diesel at $0.40 per kWh — the highest electricity cost in the world. They use approximately 30 MW peak power. Your team must design an OTEC system that can replace diesel generation, analyzing the thermodynamic, economic, and environmental feasibility of the project.

Engineering Design Thinking

1. WHERE WOULD YOU SITE THE PLATFORM RELATIVE TO THE ISLAND, AND HOW DEEP MUST THE COLD WATER PIPE EXTEND?
2. WHAT WORKING FLUID WOULD YOU USE AND WHAT IS ITS THERMODYNAMIC CYCLE EFFICIENCY?
3. HOW WOULD YOU MANAGE THE COLD DEEP WATER DISCHARGE TO MINIMIZE ENVIRONMENTAL DISRUPTION WHILE POTENTIALLY CREATING BENEFICIAL SIDE PRODUCTS?
4. WHAT IS YOUR PROJECTED COST PER KILOWATT-HOUR AND PAYBACK PERIOD COMPARED TO CONTINUED DIESEL GENERATION?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Ocean Engineers design offshore structures, marine energy systems, and underwater infrastructure. They work at renewable energy companies, offshore engineering firms, naval architecture firms, and government agencies like NOAA, earning $75,000-$140,000/year. Marine Biologists who assess the environmental impact of ocean energy projects earn $50,000-$100,000/year.